

TECHNICAL ASSESSMENT REPORT

Airtable Inventory Management & Automation

Relational Database Configuration & n8n REST API Workflow Integration

GitHub Repository: [airtable-n8n-inventory-management](#)

Tech Stack: Airtable, n8n Automation, REST API

Demo Recording: Loom Video Linked in Submission

Status: Successfully Tested & Validated

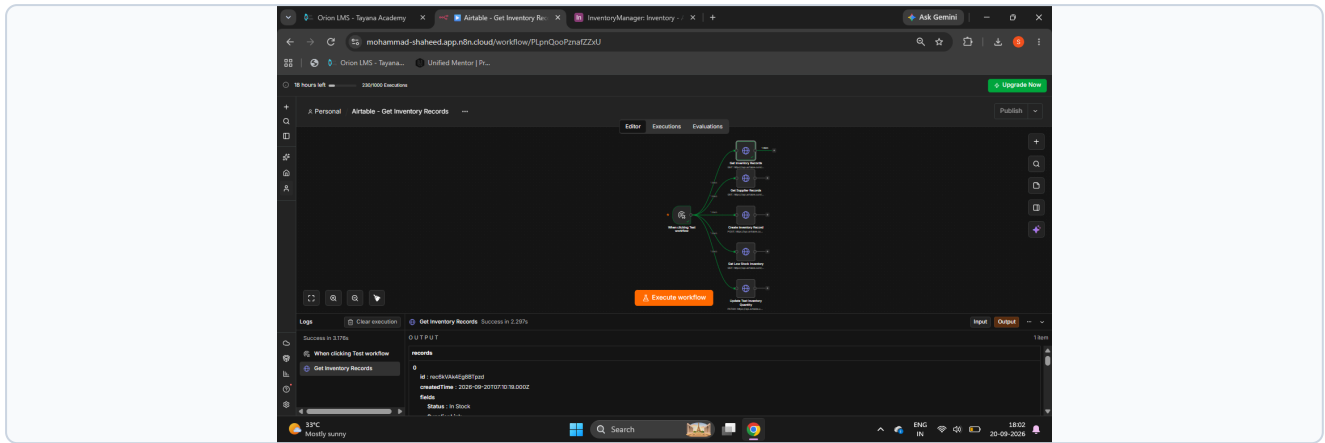
1. Project Overview & Objectives

This assessment demonstrates the implementation of a no-code relational inventory database in Airtable integrated seamlessly with n8n automated workflows using REST API endpoints, formula filters, and attachment handling.

- Configured relational database structure (`Inventory` & `Suppliers`).
- Established secure token-based API authentication (PAT).
- Automated record retrieval, creation, and conditional filtering.

2. n8n Workflow Architecture

The automated n8n workflow executes sequential and parallel API operations triggered manually to interact with Airtable endpoints.



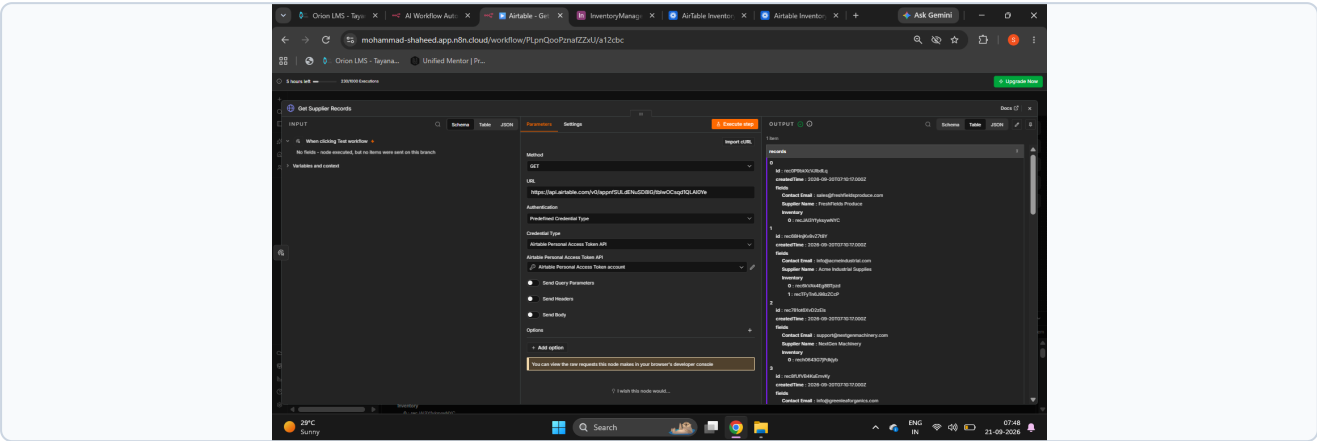
3. Airtable Database Schema

The database consists of two core linked tables managing stock details and supplier information efficiently.

Field Name	Type	Description
Product Name	Single Line Text	Name of product
Qty	Number	Available stock count
Price	Currency	Unit pricing
Product Image	Attachment	Visual representation file
Status	Single Select	Stock condition
Supplier Link	Linked Record	Associated supplier

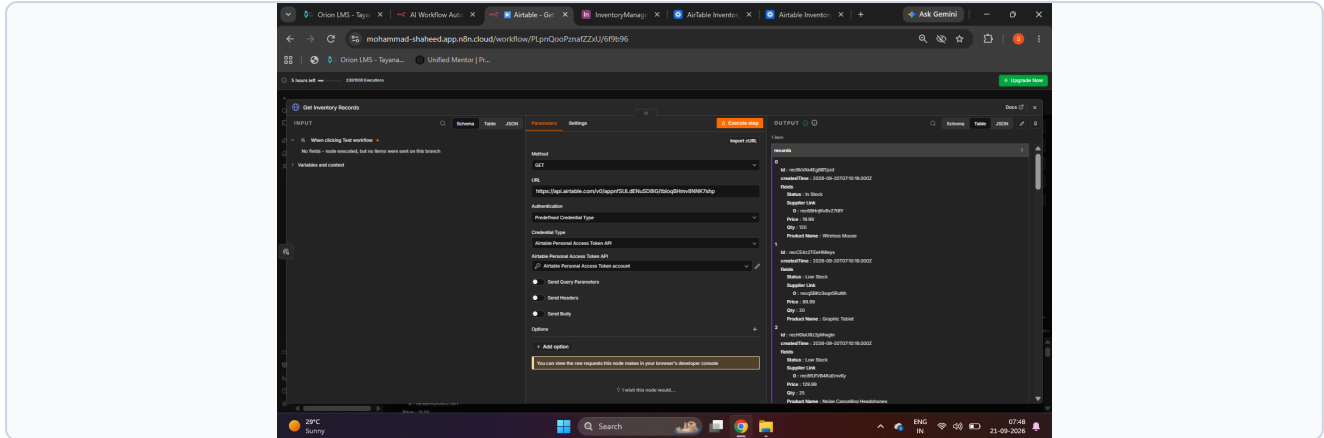
4. Supplier Records Integration

Fetching supplier records via the REST API to establish relationship links.



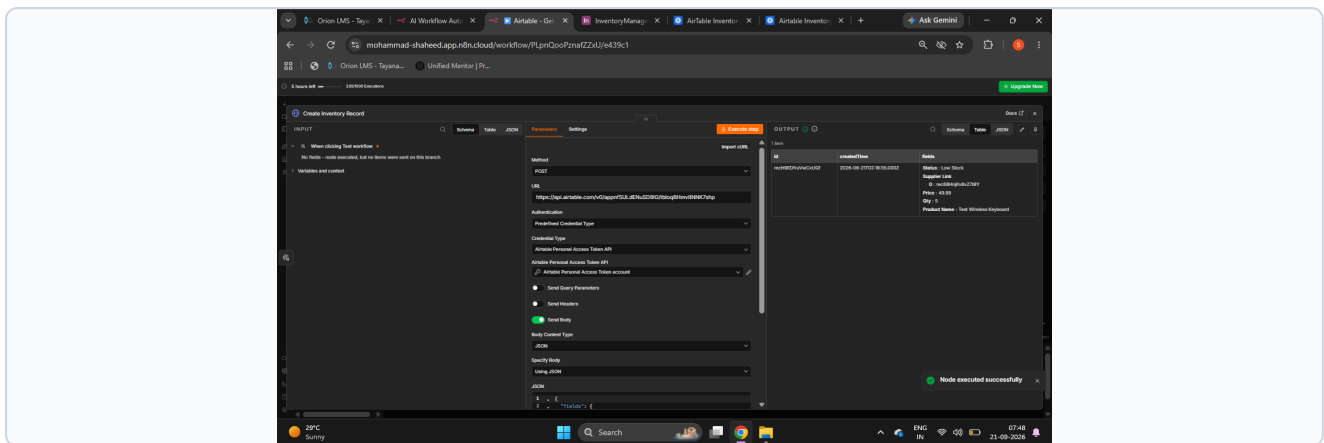
5. Inventory Record Retrieval

Executing GET requests to fetch active inventory list from Airtable.



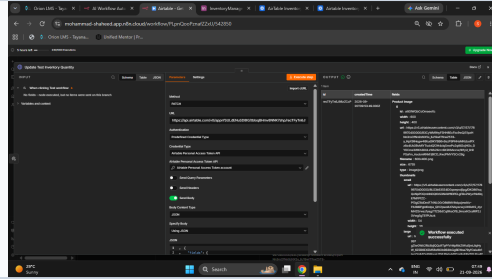
6. Programmatic Record Creation

Creating a test inventory item (`Test Wireless Keyboard`) with supplier association and initial low stock status.



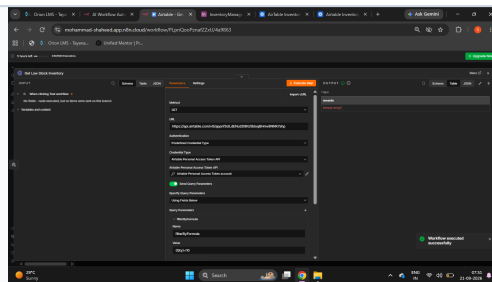
7. Low-Stock Formula Filtering

Using `filterByFormula` with expression `{Qty}<10` to isolate items below threshold.



8. Quantity Update & Validation

Updating quantity from 5 to 15 correctly returned an empty records array, verifying conditional filtering logic.



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Loom Demo: [View Demonstration Video](#)